

Course Code: CSE 3127	Course Title: Cloud Application Development Type of Course: Theory Only	L-T-P-C	3 -0	0	3
Version No.	1.0				
Course Pre-requisites	Cloud Computing Basics				
Anti-requisites	NIL				
Course Description	The Cloud Application Development Foundations Specialization program will teach students the tools and technologies that successful software developers use to build, deploy, test, run, and manage Cloud Native applications – putting them in an advantageous position to begin a new career in a highly in-demand area. The course will provide the students’ knowledge on cloud computing and related concepts, cloud services, applications developments of Amazon web services, Cloud architecture and programming model, map reducing in cloud, virtualization, applying virtualization, Cloud Resource Management and Scheduling, Cloud Security issues.				
Course Objective	The objective of the course is to familiarize the learners with the concepts of Cloud Application Development and attain Employability through Participative Learning techniques.				
Course Out Comes	On successful completion of this course the students shall be able to: Understand the Define cloud computing and related concepts and Memorize the Cloud architecture and programming model. [Comprehension] Identify compute intensive model and date intensive model and Understand the Cloud Resource Management and Scheduling. [Comprehension] Understand the Cloud Security issues and Identify the how standards deal with cloud services and virtualization. [Application] Understand the cloud resource virtualization and Identify the application virtualization, applying virtualization. [Application] Understand compliance for the cloud provider vs compliance for the customer. [Comprehension]				
Course Content:					
Module 1	INTRODUCTION AND CLOUD APPLICATION DEVELOPMENT	Assignment	Knowledge, Quizzes	No. of Classes:8	
Topics: Introduction: Definition, Characteristics, Benefits, challenges of cloud computing, cloud models: service IaaS(infrastructure as service),PaaS(platform as a service),SaaS(software as a service), deployment models-public, private, hybrid, community; Types of cloud computing: Grid					

computing utility computing, cluster; computing Cloud services: Amazon, Google, Azure, online services, open source private clouds, SLA; Applications of cloud computing: Healthcare, energy systems, transportation, manufacturing, education, government, mobile communication, application development.

Assignment: Types of cloud and their comparisons.

Module 2	CLOUD ARCHITECTURE, PROGRAMMING MODEL	Assignment	Knowledge, Quizzes	No. of Classes:7
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Topics:

Cloud Architecture, programming model: NIST reference architecture, architectural styles of cloud applications, single, multi, hybrid cloud site, redundant, non-redundant, 3 tier, multi-tier architectures; Programming model: Compute and data intensive.

Assignment: Cloud Architecture, architectural styles of cloud applications.

Module 3	CLOUD RESOURCE VIRTUALIZATION	Case Study	Application, Quizzes	No. of Classes:8
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Topics:

Cloud resource virtualization: Basics of virtualization, types of virtualization techniques, merits and demerits of virtualization, Full vs Para - virtualization, virtual machine monitor/hypervisor.

Virtual machine basics, taxonomy of virtual machines, process vs system virtual machines.

Case Study: Cloud resource virtualization: Basics of virtualization, types of virtualization techniques.

Module 4	CLOUD RESOURCE MANAGEMENT AND SCHEDULING	Case study	Application, Quizzes	No. of Classes:9
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Topics:

Cloud Resource Management and Scheduling: Policies and mechanisms for resource management, resource bundling, combinatorial, fair queuing, start time fair queuing, borrowed virtual time, cloud scheduling subject to deadlines, scheduling map reduce applications subject to deadlines, resource management and application scaling.

Case Study: Cloud Resource Management and Scheduling.

Module 5	CLOUD RESOURCE MANAGEMENT AND SCHEDULING	Case study	Application, Quizzes	No. of Classes:8
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Topics: Cloud Security: Risks, privacy and privacy impacts assessments; Multi-tenancy issues, security in VM, OS, virtualization system security issues and vulnerabilities; Virtualization system-specific attacks: Technologies for virtualization-based security enhancement, legal. Case Study: Cloud Security: Risks, privacy and privacy impacts assessments.
Targeted Application & Tools that can be used: Public cloud platforms like AWS, GCP and Azure.
Project work/Assignment: Create an Amazon EC2 Instance (Linux) or use equivalent other cloud platform such as Google Cloud or Azure to create a virtual machine service. Create an Amazon S3 Bucket or use equivalent other cloud platform such as Google Cloud or Azure to create a storage service. Create a static website in AWS using S3 and cloud front.
Textbook(s): Dan Marinescu, "Cloud Computing: Theory and Practicell", M K Publishers, 1st Edition, 2013, Kai Hwang, Jack Dongarra, Geoffrey Fox," Distributed and Cloud Computing, From Parallel Processing to the Internet of Thingsll", M K Publishers, 1st Edition, 2011.
References Anthony T. Velte, Toby J. Velte, Robert Elsenpeter, "Cloud Computing: A Practical Approach", McGraw Hill, 1st Edition, 2009. Arshdeep Bahga, "Cloud Computing: A Hands on Approach", Vijay Madisetti Universities Publications, 1 st Edition, 2013.
Web Resources and Research Articles: https://www.oracle.com/in/cloud/application-development http://computingcareers.acm.org/?page_id=12 http://en.wikibooks.org/wiki/cloud application http://www.acadmix.com/eBooks_Download http://www.ibm.com pu.informatics.global, https://sm-nitk.vlabs.ac.in/
Topics relevant to "EMPLOYABILITY SKILLS": EC2 for developing Employability Skills through Participative Learning techniques. This is attained through assessment component mentioned in course handout.

